

Eventual Sums of Squares under Heat Flow: The Coercive-Leading-Form Case

Candidate partial result for arXiv:2211.04416

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Status: candidate partial result, likely valid pending human review. The source problem remains open for degenerate leading forms.

1 Source problem

Curto, di Dio, Korda, and Magron study the action of the polynomial heat semigroup on nonnegative polynomials [1]. On PDF page 26 they ask:

Open Problem 5.1. Let $f \in \text{Pos}(\mathbb{R}^2) \setminus \text{SOS}(\mathbb{R}^2)$. Is it true that there is always a $T = T(f) > 0$ such that $e^{t\Delta}f \in \text{SOS}(\mathbb{R}^2)$ for all $t \geq T$?

The previous example demonstrates why we have to restrict to $g \in C_b^\infty$.

5. Summary and open questions

We investigated time-dependent moments from partial differential equations. At first we gave the dual action on the polynomials (Theorem 2.2). While for general coefficients g and h we leave $\mathbb{R}[x_1, \dots, x_n]$ in general, for coefficients independent on x we remain in $\mathbb{R}[x_1, \dots, x_n]$. We investigated the action of the heat equation on non-negative polynomials more deeply. In Examples 3.8, 3.11 and 3.13 to 3.15 we have collected examples of non-negative polynomials on $\mathbb{R}^2$ which are not sums of squares. But under the action of the heat equation after finite time they become sums of squares. The highest degree part is a non-negative homogeneous polynomial in two variables and hence a sum of squares [59]. If this holds for all bivariate non-negative polynomials is still open.

Open Problem 5.1. Let $f \in \text{Pos}(\mathbb{R}^2) \setminus \text{SOS}(\mathbb{R}^2)$. Is it true that there is always a $T = T(f) > 0$ such that $\mathbf{p}_f(x, t) \in \text{SOS}(\mathbb{R}^2)$ for all $t \geq T$?

Figure 1: Open Problem 5.1 and its context in arXiv:2211.04416, PDF page 26.

For $m, d \geq 1$, let $\mathcal{H}_{m,d}$ be the real vector space of degree- d forms in m variables, and let $\Sigma_{m,2d}$ be the cone of sums of squares of forms in $\mathcal{H}_{m,d}$. A form is a *strict sum of squares* if it lies in the interior of $\Sigma_{m,2d}$; equivalently, it has a positive-definite Gram matrix in a monomial basis of $\mathcal{H}_{m,d}$.

2 Proof intuition

Parabolic scaling separates the highest homogeneous part p of f from all lower-order terms. After homogenization in a new variable z , the normalized late-time heat orbit converges to $e^{z^2\Delta}p$.

If p is a strict SOS, choose a Gram factorization $p = \sum_j q_j^2$ in which the q_j form a basis of all degree- d forms. The Gaussian formula for heat flow writes the limit as an average of the squares of the translated forms $q_j(x + \sqrt{2}zG)$. Those translations span every degree- d form in the old variables together with z . Consequently the averaged Gram matrix is positive definite: the limiting form is an *interior* SOS point, not merely an SOS point. Small lower-degree perturbations are therefore absorbed for all sufficiently large times.

For two variables, every nonnegative binary form is SOS by Hilbert's binary form theorem [2]. A positive-definite binary form is an interior point of that cone, so the argument covers exactly the source problem's coercive-leading-form case. The remaining difficulty is the boundary case where the leading form vanishes in a real projective direction.

3 A Gaussian interior-lifting lemma

Write $x = (x_1, \dots, x_n)$ and let $\Delta_x = \sum_{i=1}^n \partial_{x_i}^2$.

Lemma 1 (Gaussian lift preserves strict SOS). *If $p \in \mathcal{H}_{n,2d}$ is a strict sum of squares, then*

$$P(x, z) := e^{z^2 \Delta_x} p(x)$$

is a strict sum of squares in $\mathcal{H}_{n+1,2d}$.

Proof. Let $v(x)$ be the monomial vector of $\mathcal{H}_{n,d}$. Strict SOS membership gives

$$p(x) = v(x)^\top G v(x), \quad G \succ 0.$$

Factoring G shows that

$$p = \sum_{j=1}^N q_j^2, \quad N = \dim \mathcal{H}_{n,d},$$

where $q_1, \dots, q_N$ form a basis of $\mathcal{H}_{n,d}$.

Let γ be standard Gaussian measure on $\mathbb{R}^n$. The Gaussian moment identity gives

$$\begin{aligned} P(x, z) &= \int_{\mathbb{R}^n} p(x + \sqrt{2}z\xi) d\gamma(\xi) \\ &= \int_{\mathbb{R}^n} \sum_{j=1}^N q_j(x + \sqrt{2}z\xi)^2 d\gamma(\xi). \end{aligned} \tag{1}$$

Every integrand in (1) is the square of a form in $\mathcal{H}_{n+1,d}$. Expanding its coefficient vector and integrating its rank-one Gram matrix therefore produces a positive-semidefinite Gram matrix A for P .

It remains to show $A \succ 0$. The coefficient vectors of

$$q_j(x + z\xi), \quad 1 \leq j \leq N, \text{quad } \xi \in \mathbb{R}^n, \tag{2}$$

span $\mathcal{H}_{n+1,d}$. Since the q_j form a basis, it suffices to use all monomials of degree d in place of the q_j . Given a monomial $x^\alpha z^k$ with $|\alpha| + k = d$, choose a multi-index β with $|\beta| = k$. Differentiating the translate of $x^{\alpha+\beta}$ gives

$$\partial_\xi^\beta (x + z\xi)^{\alpha+\beta} \Big|_{\xi=0} = \frac{(\alpha + \beta)!}{\alpha!} x^\alpha z^k. \tag{3}$$

The coefficients of a polynomial map lie in the span of its values, so (3) proves the spanning claim. Thus a vector in the kernel of A would be orthogonal to a spanning set and must vanish. Hence $A \succ 0$, proving that P is a strict SOS. $\square$

4 Eventual SOS theorem

Theorem 2. *Let $f \in \mathbb{R}[x_1, \dots, x_n]$ have degree $2d$, and let $p = f_{2d}$ be its highest homogeneous part. If p is a strict sum of squares, then there is $T > 0$ such that*

$$e^{t\Delta_x} f \in \text{SOS}(\mathbb{R}^n) \quad \text{for every } t \geq T.$$

No nonnegativity assumption on f is required.

Proof. Decompose $f = \sum_{k=0}^{2d} f_k$ into homogeneous parts. For $t > 0$ put $s = t^{-1/2}$. Homogeneity and the finite heat-series expansion give

$$t^{-d}(e^{t\Delta_x} f)(\sqrt{t}x) = \sum_{k=0}^{2d} s^{2d-k} e^{\Delta_x} f_k(x). \quad (4)$$

The degree- $2d$ homogenization of the right-hand side is

$$Q_s(x, z) = \sum_{k=0}^{2d} s^{2d-k} z^{2d-k} e^{z^2 \Delta_x} f_k(x). \quad (5)$$

Consequently, as $s \downarrow 0$,

$$Q_s \longrightarrow Q_0 = e^{z^2 \Delta_x} p \quad \text{coefficientwise in } \mathcal{H}_{n+1, 2d}. \quad (6)$$

Lemma 1 places Q_0 in the interior of the finite-dimensional closed cone $\Sigma_{n+1, 2d}$. This interior is open, so (6) implies that $Q_s \in \Sigma_{n+1, 2d}$ for every sufficiently small positive s . Dehomogenizing (5) at $z = 1$ and using (4) shows that the normalized polynomial in (4) is SOS. Positive scalar multiplication and the invertible scaling $x \mapsto \sqrt{t}x$ preserve SOS membership. Thus $e^{t\Delta_x} f$ is SOS for every sufficiently large t . $\square$

Corollary 3 (Coercive bivariate case). *Let $f \in \mathbb{R}[x, y]$ have degree $2d$ and leading form p . If*

$$p(u) > 0 \quad (u \in \mathbb{R}^2 \setminus \{0\}),$$

then $e^{t\Delta} f$ is SOS for all sufficiently large t . In particular, Open Problem 5.1 has an affirmative answer whenever the nonnegative input polynomial has a positive-definite leading form.

Proof. Positive-definite binary forms are precisely the interior points of the cone of nonnegative binary forms. Hilbert's binary-form theorem identifies that cone with $\Sigma_{2, 2d}$ [2]. Hence p is a strict SOS, and Theorem 2 applies. $\square$

5 Limitations and boundary obstruction

Corollary 3 does not settle the full source problem. If the leading binary form p has a real projective zero, then p lies on the boundary of the binary SOS cone, and $e^{z^2 \Delta} p$ lies on a proper face of the ternary SOS cone. Coefficient convergence to a boundary point gives no neighborhood of SOS forms. Moreover, lower homogeneous pieces can decrease the multiplicity at an infinity zero, so the normalized path can move through nested faces rather than remain in a single relative interior. This is the specific obstruction left by the proof.

The result nevertheless removes the generic case: among nonnegative binary leading forms of fixed degree, positive-definite forms constitute the interior of the cone. Only the locus with a real projective zero remains.

6 Novelty check and verification

The source problem and notation were checked against arXiv:2211.04416, Open Problem 5.1 on PDF page 26. A bounded search through 13 August 2026 covered the local run indexes, the parsed arXiv corpus, exact problem wording, the source title and arXiv id, and combinations of “heat”, “eventual sum of squares”, “coercive polynomial”, and “leading homogeneous form”. The later arXiv:2506.16321 cites the source but does not answer this heat-flow problem. No explicit prior statement of Theorem 2 or Corollary 3 was located. This is bounded evidence, not a claim of exhaustive novelty.

The proof is exact and has no computational dependency. Verification checked the powers of t in (4), the homogenization exponents in (5), the Gaussian variance in (1), and the monomial-spanning identity (3). Human review should focus on those four points and on the equivalence between strict SOS membership and a positive-definite Gram matrix.

References

- [1] R. E. Curto, P. J. di Dio, M. Korda, and V. Magron, *Time-dependent moments from partial differential equations and the time-dependent set of atoms*, arXiv:2211.04416 (2022; revised 2023).
- [2] D. Hilbert, *Ueber die Darstellung definiter Formen als Summe von Formenquadraten*, *Mathematische Annalen* **32** (1888), 342–350.